CASCADING FAILURES IN POWER GRIDS

A replication study based on Schäfer et al. (2018)

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Motivation

Why is it important?

- Power grids are critical infrastructure
- Failures can affect millions of people

Problem: Cascading Failures

- Small disturbances can trigger large-scale outages
- Domino effect due to load redistribution

Limitation of existing models

- Many approaches rely on static assumptions
- underestimate systemic risk

Static vs. Dynamic Models

Static Models

- Assume steady-state power flows
- Failures occur only if capacity is exceeded
- No time dimension

→ only final equilibrium

Dynamic Models

- Capture transient system dynamics
- Include oscillations in frequency and flows
- Failures can occur during adjustment

→ full temporal evolution

Model Idea

Power grid as network →

- Nodes = Generators & Consumers
- Edges = Transmission Lines

Dynamic Modelling →

- each node is a coupled oscillator
- phase angle + frequency

Swing Equation →

- Captures synchronization dynamics of grid
- System responds to disturbances

Failure Mechanism I

$$F_{ij}(t) = K_{ij} \sin(\theta_j(t) - \theta_i(t))$$

- F_{ij} – Power Flow
  - how much power is transmitted along a line
- K_{ij} – Coupling strength
  - maximum transmission capacity
- θ_i – Phase Angle

→ flow depends on phase differences between nodes

Failure Mechanism II

$$|F_{ij}(t)| > C_{ij} = \alpha K_{ij}$$

- α - tolerance parameter (safety margin)

Step 1

Initial failure

- one transmission line is removed

Step 2

Load redistirbution

- power flows are rerouted across remaining lines

Step 3

Secondary failures

- other lines may become overloeaded

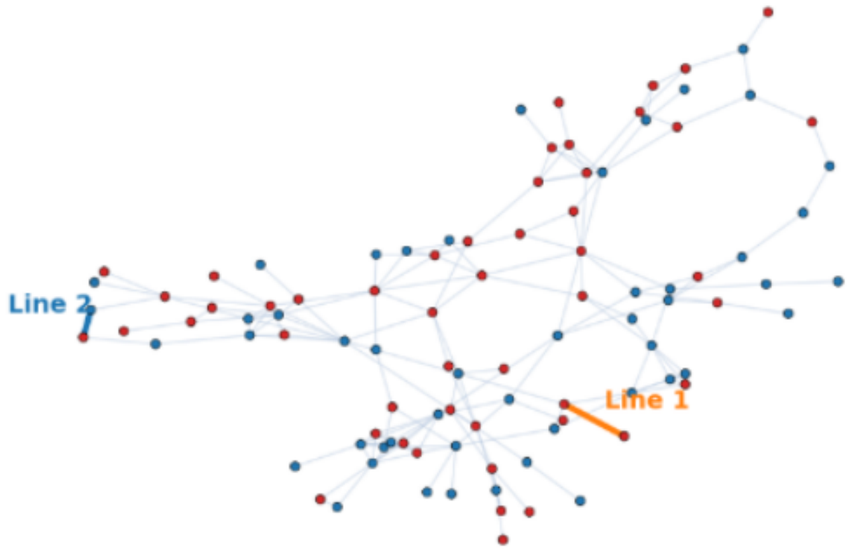
Step 4

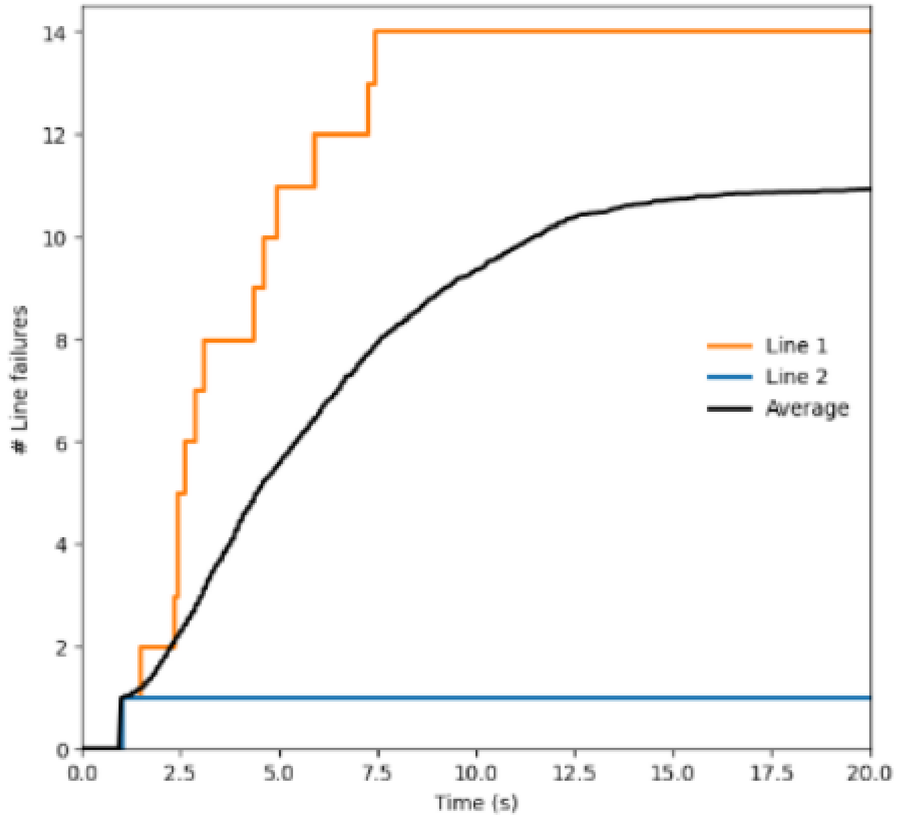
Process repeats → cascade emerges

Key Results I

- Strong dependence on initial failure
  - Line 1 → large cascade (~14 failures)
  - Line 2 → almost no cascade (~1 failure)

- Cascades develop over time
  - step-by-step increase
  - most occur within first seconds





Key Results II

Distributed Power

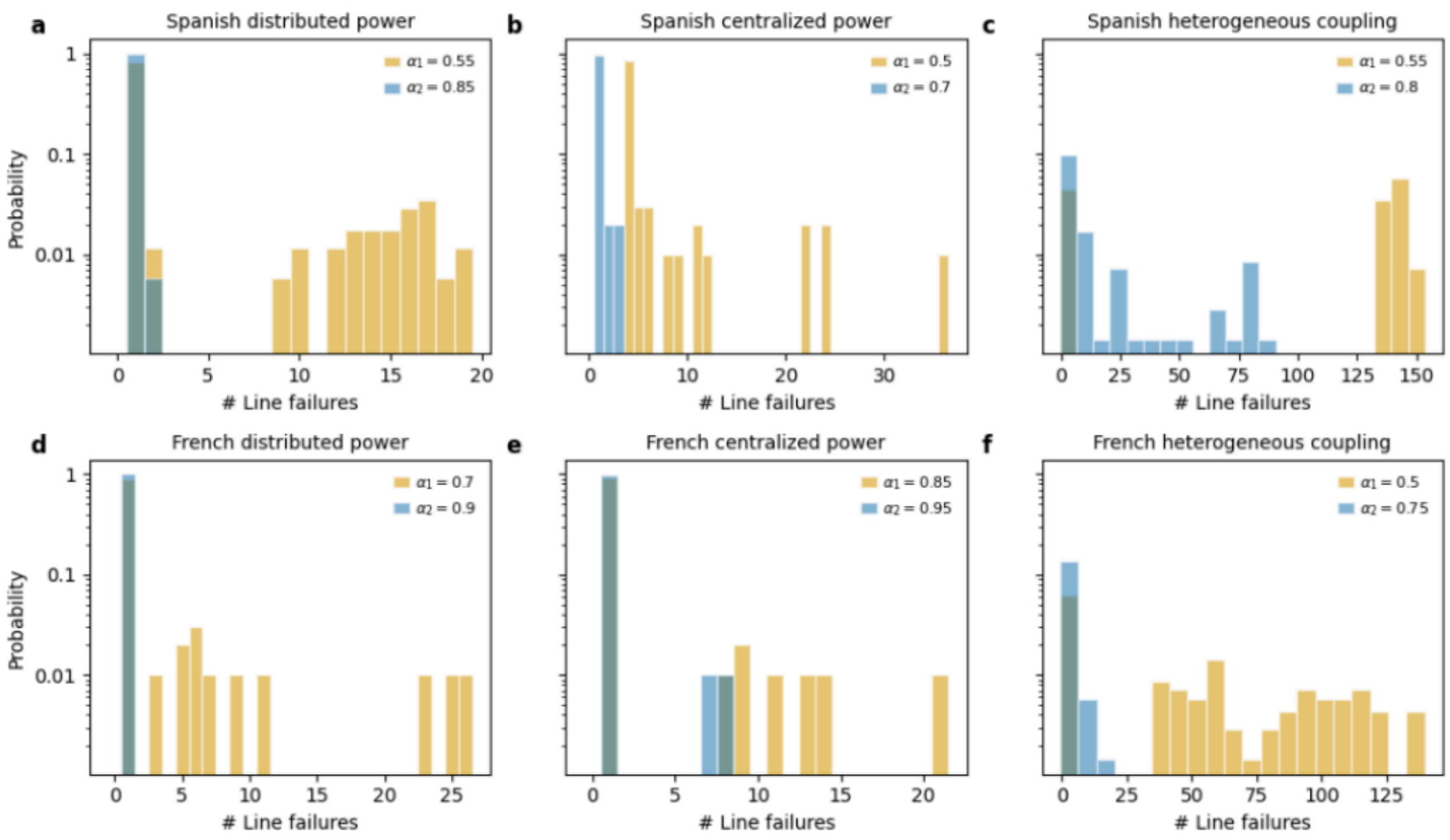
- many small producers
- load is evenly distributed

Centralized Power

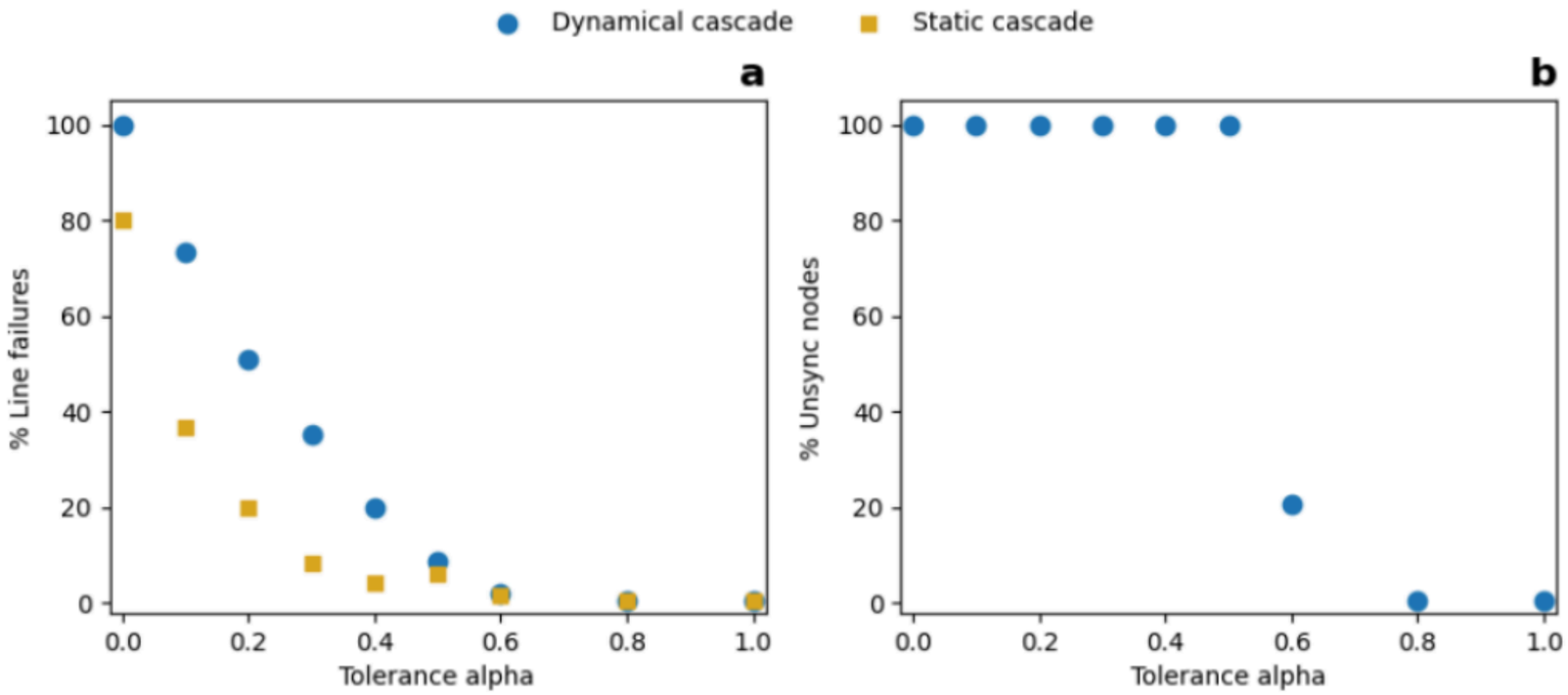
- few large power plants
- higher load on specific transmission paths

Heterogenous Coupling

- some lines are stronger or weaker than others
- uneven load distribution



Key Results III



Dynamic cascades are much larger → static models underestimate failures

higher tolerance → fewer failures and less desynchronisation

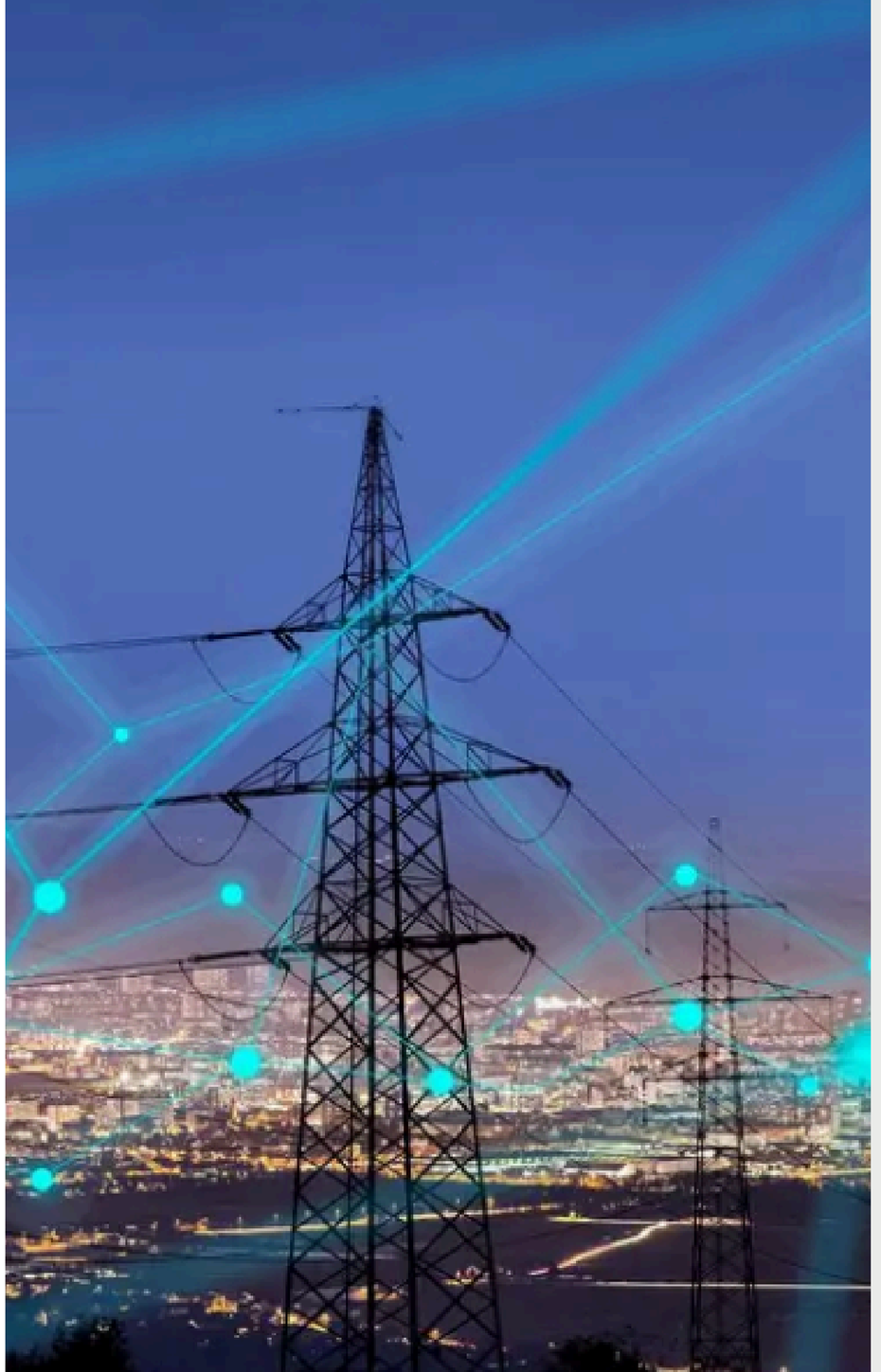
Limitations

Computational constraints

- Analysis based on reduced sample of initial failures

Missing real-world mechanisms

- No protection systems (e.g. relays, load shedding)

Differences to author's code

- simplified failure detection (discrete vs. continuous detection)
- immediate line removals (immediate vs. persistent removal)